

Meerkat-TRIZ: A Fine-Tuned Language Model for TRIZ-Based Inventive Problem Solving with Dual-Track Evaluation and Cross-Family Judge Verification

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Model: <https://huggingface.co/Meerkat-AI/Meerkat-TRIZ-v1>
Code & benchmark: <https://github.com/coidea-sys/meerkat-triz>

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Abstract

TRIZ (Theory of Inventive Problem Solving) is the most widely used systematic innovation methodology in engineering practice, yet existing work combining TRIZ with large language models (LLMs) has relied exclusively on prompting, retrieval, or multi-agent orchestration of general-purpose models. We present Meerkat-TRIZ, to our knowledge the first publicly released TRIZ-domain *fine-tuned* LLM: a QLoRA adapter (0.24% of parameters) trained on a Qwen3.6-35B-A3B MoE base with an 11k-example curriculum covering six TRIZ task families plus safety refusals. Because domain claims are only as strong as their measurement, the paper’s second contribution is an evaluation methodology designed to survive adversarial scrutiny: (i) **TRIZ-Bench**, a 300-item six-task benchmark whose questions and keyword sets are public; (ii) a *dual-track* protocol pairing an objective keyword track with an anti-verbosity LLM-judge track; (iii) paired-bootstrap significance testing with fixed seeds throughout; (iv) a *cross-family judge panel* with family recusal and a circuit breaker that freezes shipping when an in-family judge contradicts the external consensus; and (v) an eight-gate decision procedure (G0–G8) that has repeatedly caught regressions that single-metric leaderboards would have missed. Applying this apparatus, we show that Meerkat-TRIZ v1 improves over its 35B base by +0.39 (± 0.10 , 95% CI) on the in-family judge but only +0.09 to +0.10 under two external judges and -0.05 under a third—a $\sim 4\times$ inflation pattern that would have gone undetected with a single judge. An *equal-length control* then isolates the residual confound: when the base is instructed to match the fine-tune’s answer length, the in-family gain vanishes (-0.07 , n.s.) while the base’s own in-family score rises by +0.46 from brevity alone—yet under two of three *external* judges the same length-matched base falls *below* the fine-tune by +0.19 to +0.27 (significant). Length was masking a genuine content-quality gain: the fine-tune is better at a level length budget, and at natural lengths delivers that content at $0.4\times$ the answer length with keyword coverage the base cannot retain when compressed. On the objective keyword track, the fine-tune matches its base (± 0.0001) and both outscore GPT-5.4, Claude Sonnet 4.6, Claude Opus 4.8 and Gemini 3.5 Flash on TRIZ keyword coverage; on the judge track the best frontier models still lead, with Claude Opus 4.8 scoring significantly *below* the 35B base. A 120-item

general-ability probe shows no significant capability sacrifice from domain fine-tuning (−1.25pp vs. predecessor, floor −5pp). We release the model, the evaluation harness, the public benchmark, and all judge caches to make every number in this paper recomputable.

Introduction

TRIZ (*Teoriya Resheniya Izobretatelskikh Zadach*, Theory of Inventive Problem Solving), developed by Altshuller from the analysis of hundreds of thousands of patents, codifies invention as the resolution of *contradictions*: improving one engineering parameter while avoiding the degradation of another, using 40 inventive principles, a contradiction matrix, separation principles, and the ARIZ algorithm. Despite its industrial adoption, TRIZ remains expertise-bound: mapping a messy real-world problem onto the correct standard contradiction, and then committing to the right principles with defensible justification, is precisely the step practitioners find hardest.

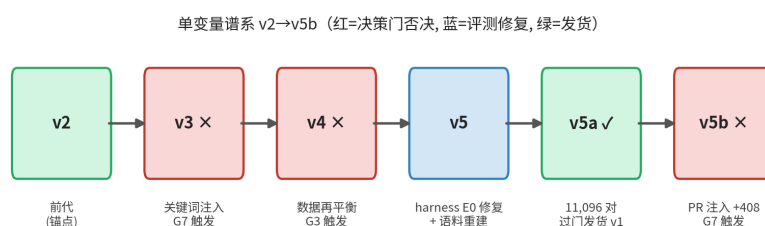
Large language models are a natural vehicle for lowering this barrier, and several groups have explored the combination . All of these works, however, treat the LLM as a general-purpose reasoner steered by prompts, retrieval, or agent pipelines. None trains the model on the domain, and none evaluates with the statistical machinery that the LLM-as-judge literature has shown to be necessary . Two questions therefore remain open: (1) does domain fine-tuning actually move TRIZ task quality beyond what a strong base model or a frontier model already achieves? and (2) how should such a claim be measured when the most convenient instrument—an LLM judge from the same model family that generated the training data—is itself suspect?

This paper answers both. Our contributions are:

1. **The first public TRIZ fine-tuned LLM.** Meerkat-TRIZ v1 is a QLoRA adapter on Qwen3.6-35B-A3B (MoE, 256 routed experts; adapters on 12 shared projections, 0.24% of parameters), trained on an 11k-example, seven-subset curriculum. Model, training configuration, and data-construction pipeline are public.
2. **A measurement-first evaluation methodology.** Dual-track scoring (objective keyword track + anti-verbosity judge track), paired-bootstrap inference with fixed seeds, a 120-item general-ability probe, and an eight-gate shipping decision procedure (G0–G8) whose gates are machine-checkable and whose backtests reproduce our past ship/no-ship decisions exactly.
3. **Cross-family judge verification.** A panel of external judges from three vendors, with family recusal (a judge never scores its own family’s generations) and a circuit breaker (G8) that blocks shipping when in-family and external verdicts contradict. We quantify the resulting correction: the in-family judge’s headline effect (+0.39) deflates to +0.09/+0.10/−0.05 under external judges, while external judges agree with each other at Spearman 0.63–0.75 but only 0.27–0.31 with the in-family judge. An equal-length generation control completes the correction—and then reverses it upward: once the base is instructed to match the fine-tune’s answer length, the in-family effect vanishes (−0.07, n.s.), exposing

the headline as predominantly a length artifact, while two of three external judges score the fine-tune significantly *above* the length-matched base (+0.19/+0.27), isolating a genuine content-quality gain that length had masked.

4. **A frontier baseline.** TRIZ-Bench results for GPT-5.4, Claude Sonnet 4.6, Claude Opus 4.8, and Gemini 3.5 Flash under the identical protocol, with per-judge disclosure of every number.
5. **A documented failure-driven iteration lineage.** Five model generations (v2→v5b), each changed by a single variable and each decided by the gates, including three rejected generations and the attribution analyses that rejected them—evidence that the methodology, not luck, produces the improvement (Figure 1).



Single-variable lineage v2→v5b. Each generation changes exactly one factor (annotated below); red generations were rejected by the decision gates, blue repaired the measurement harness, and green shipped as Meerkat-TRIZ v1.

Section 2 positions against prior TRIZ×LLM work and the LLM-as-judge literature. Section 3 describes data and training. Section 4 defines the evaluation apparatus. Section 5 reports results, including the frontier leaderboard. Section 6 analyzes failure modes and judge bias. Section 7 is an unusually detailed limitations section; Section 8 concludes.

Related Work

TRIZ meets LLMs: prompting, not training.

AutoTRIZ automates TRIZ ideation by prompt-structured LLM calls and evaluates on textbook cases; TRIZ-GPT compares prompt strategies for GPT-3.5/4 inside a TRIZ workflow; TRIZ Agents decomposes the methodology across a multi-agent pipeline; Gusarov frames TRIZ as a reasoning scaffold for materials science with local models. Common traits: the model weights are never touched; evaluation is by case study or heuristic LLM scoring without significance testing; and no public benchmark exists. Our work is complementary in spirit—we keep TRIZ’s structured problem formulation as the task definition—but orthogonal in mechanism: domain adaptation through supervised fine-tuning, with measurement treated as a first-class deliverable.

Domain fine-tuning.

Parameter-efficient fine-tuning has made single-GPU domain adaptation routine. What remains rare is discipline in what is claimed from it. Our training recipe is deliberately plain (QLoRA, fixed hyperparameters, two-epoch cosine horizon); the novelty is not the optimizer but the decision procedure around it: every generation changes one variable and must pass eight machine-checkable gates.

LLM-as-judge and its biases.

Zheng et al. established LLM judging at scale and documented position and verbosity biases; Panickssery et al. showed evaluators favor their own generations; length-controlled variants of automatic arenas exist precisely because judges reward length. Our design imports these lessons: an anti-verbosity clause in the rubric, position-swap validation in harness development, fixed-temperature scoring, and—most importantly—treating any single judge, especially one from the data generator’s family, as a hypothesis to be falsified by external judges rather than as ground truth.

Benchmarks and contamination.

Holistic multi-metric evaluation , live refreshed benchmarks , and public reproducible harnesses motivate our release shape: public questions and keyword sets, withheld reference answers, documented decontamination (character 3-gram Jaccard, threshold 0.5, with a 0.4–0.5 human-review band), versioned snapshots, and full judge-cache disclosure so that every table cell can be recomputed independently.

Model and Training

Base model and adapter

Meerkat-TRIZ v1 attaches QLoRA adapters ($r=64$, $\alpha=128$, dropout 0, RSLoRA off) to twelve shared projection modules (`q`, `k`, `v`, `o_proj`, `in_proj_qkv`, `z`, `b`, `a`, `out_proj`, `gate`, `up`, `down_proj`) of Qwen3.6-35B-A3B , a Mixture-of-Experts base with 256 routed experts. Only 0.24% of parameters are trainable; the routed experts themselves are untouched. Training is supervised fine-tuning in pure BF16 on a single unified-memory accelerator, batch $1 \times \text{grad-accum } 8$, $\text{lr } 2 \times 10^{-4}$ with a cosine schedule whose horizon is two epochs (four-epoch cap as safety), max sequence 2048, completion-only loss, seed 42. Hyperparameters were fixed by a small sweep at an earlier generation and then *frozen*: every subsequent generation changes data only, preserving single-variable comparability across the lineage (Table 1).

Training data

The v5-line corpus derives from a cleaned collection of Chinese TRIZ textbooks, courseware, and proceedings, expanded by model-generated long-form answers under hard quality gates (length bands, tail-template blacklists, dedup) and decontaminated against the evaluation gold set by character 3-gram Jaccard (≥ 0.5 drop; 0.4–0.5 human review). The final v5b training set contains 11,421 prompt–completion pairs in seven subsets: concept explanation (2,996), innovation assessment (3,037), case generation (2,857), ARIZ guidance (1,442), contradiction analysis (313), principle recommendation (521), and safety refusal (255).

The principle-recommendation subset deserves note because it is the paper’s running case study (Section 6). Attribution on the previous generation found the subset to be the smallest (196 items), 72% of it “concise-mode” items capped at 300 characters, and—most damning—42% mislabeled (ARIZ resource questions, standard-solution questions, and patent-analysis questions carrying a principle-recommendation tag). The v5b generation audits out the 83 mislabeled items and injects 408 newly generated examples whose answers must commit to specific numbered principles with justification, pass a hedging filter (“may involve/one may consider” phrasing rejected), cover all 40 inventive principles without omission, and pass gold-set decontamination.

Model lineage. Each row changes exactly one variable relative to its predecessor; ship/reject verdicts are produced by the G0–G8 gates (Section 4.5).

Generation	Change	Gate verdict	Note
v2	v5-line corpus rebuild	ship	baseline predecessor
v3	keyword-rich data push	reject	judge track regressed
v4	data rebalance	reject	concept-explanation –8 kw
v5	new gold + corpus	superseded	training-protocol defect
v5a	protocol fix (E0)	ship (as v1)	released model
v5b	PR audit + injection	reject	G7 fired $\times 2$; no external confirmation

Evaluation Methodology

TRIZ-Bench

The benchmark contains 300 Chinese items in six task families (concept explanation, contradiction analysis, principle recommendation, ARIZ guidance, case generation, innovation assessment; 30–60 items per family), each with a question, a reference answer, and an expected keyword set. Questions and keyword sets are public (CC-

BY-NC); reference answers are withheld to slow benchmark decay. Generation protocol for all models is identical: one TRIZ-expert system message, max 2048 new tokens, deterministic decoding where the API allows.

Dual-track scoring

Keyword track (objective).

Per-item hit rate of expected keywords under a curated alias map (substring plus confirmed synonyms), fully deterministic and recomputable by anyone from the public files.

Judge track (semantic), Arm A.

An LLM judge scores each answer 0–4 on five dimensions (accuracy, completeness, TRIZ correctness, structure, overall) against the reference answer, with an explicit anti-verbosity clause (“length is not evidence of quality”), full untruncated input, temperature 0, five items per request, batch parse failures degrading to single-item retries, and a fixed judge system prompt. An Arm B variant judging only same-length-bucket pairs exists for length-confound control, but with only 17/300 qualifying pairs it is underpowered; we report Arm A and treat the length confound as an open measurement debt (Section 7).

Inference

All comparisons are paired at item level: paired bootstrap on per-item differences (10,000 resamples, seed 42), McNemar’s exact test for pass-rate asymmetries, Wilson intervals for rates, and Spearman’s ρ for judge–judge agreement. No claim in this paper is made without a confidence interval; “significant” always means a 95% CI excluding zero.

Cross-family judge panel and recusal

Every arm is scored by the in-family judge (Moonshot) and by external judges from three vendors (Claude, GPT, Gemini families). Two rules make the panel adversarial: *family recusal*—a judge never scores generations of its own family—and *full per-judge disclosure*—no averaging across judges, because averaging hides exactly the disagreement we want to inspect.

Decision gates G0–G8

Shipping is decided by eight machine-checkable gates applied to the paired statistics: G0 over-refusal $\leq 2\%$; G1 vs-base judge CI lower bound > -0.15 ; G2 vs-predecessor CI lower bound > -0.05 with at least one track significantly positive; G3 keyword-regression attribution (distinguishing true regressions from scoring artifacts); G4 small-cell ($n < 30$) subsets are descriptive only; G5 ARIZ rubric

precedence; G6 general-ability probe degradation $> -5\text{pp}$; G7 dual-track contradiction circuit breaker (tracks significantly opposing); G8 cross-family contradiction circuit breaker (in-family significantly positive while all or a majority of external judges are significantly negative $\Rightarrow$ fail). Any FAIL keeps the predecessor; any SKIP freezes the release pending data. Backtests of the gate suite reproduce all historical ship/reject decisions, including the two rejections in Table 1.

General-ability probe

A 120-item six-category general Chinese probe (no TRIZ content) is run on base, predecessor, and candidate under a neutral system prompt, to detect capability sacrifice from domain tuning.

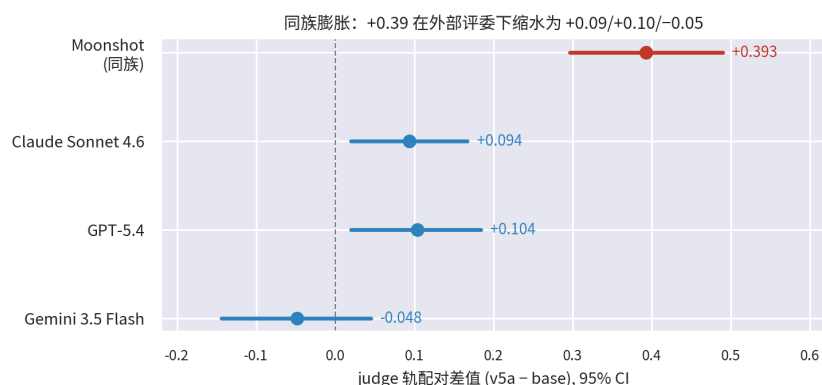
Results

Fine-tune vs. base: the judge you choose changes the answer

Table 2 is the paper’s central table. Under the in-family judge, Meerkat-TRIZ v1 (v5a) improves over base by $+0.393$ [$+0.297$, $+0.490$] on the 0–4 scale—a headline any single-judge paper would print. Under external judges the same paired differences are $+0.094$ [$+0.020$, $+0.167$] (Claude), $+0.104$ [$+0.020$, $+0.184$] (GPT), and -0.048 [-0.144 , $+0.045$] (Gemini): two significantly positive but $\sim 4\times$ smaller, one indistinguishable from zero. The direction survives verification; the magnitude does not (Figure 2). Even this deflated pattern has a residual length confound—base answers are much longer than the fine-tune’s—and Section 5.5 shows that under an equal-length control the in-family difference disappears entirely, while two of three *external* judges on the same matched arm find the fine-tune significantly *better*; we restate the fine-tune’s benefit accordingly. Meanwhile the objective keyword track is an exact tie (0.6384 vs 0.6383 , diff -0.0001 n.s.), so we report the honest composite: the fine-tune adds judged quality without adding keyword coverage, and the size of the quality gain depends on who judges—and, under the in-family judge, on how long the answers are.

Meerkat-TRIZ v1 (v5a) vs. base on TRIZ-Bench (300 items). Judge cells: paired diff [95% CI] from 10,000 paired-bootstraps, seed 42. *CI excludes zero.

Track / judge	diff (v5a — base)	sig.
Keyword (objective)	-0.0001	n.s.
Moonshot (in-family)	$+0.393$ [$+0.297$, $+0.490$]	*
Claude Sonnet 4.6	$+0.094$ [$+0.020$, $+0.167$]	*
GPT-5.4	$+0.104$ [$+0.020$, $+0.184$]	*
Gemini 3.5 Flash	-0.048 [-0.144 , $+0.045$]	n.s.

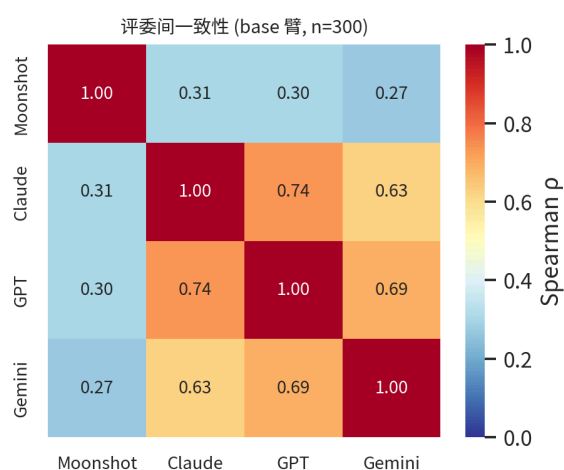


The in-family inflation pattern at a glance. Paired judge-track differences (v5a — base) with 95% CIs under each judge. The in-family judge’s headline +0.393 deflates to +0.094/+0.104/ −0.048 under three external judges—the direction survives, the magnitude does not.

Judge–judge agreement explains why the correction matters (Table 3, Figure 3). External judges cohere with each other (ρ 0.63–0.75) at roughly the level MT-Bench reports for GPT-4-vs-human agreement, but barely cohere with the in-family judge (ρ 0.27–0.31). The in-family judge is not merely more generous; it is measuring something related but different—consistent with self-preference findings , and the reason G8 exists.

Judge–judge Spearman ρ on the base arm (300 items).

Pair	ρ
Claude ~ GPT	0.738
GPT ~ Gemini	0.692
Claude ~ Gemini	0.628
Moonshot ~ any external	0.27–0.31



Judge–judge Spearman ρ on the base arm ($n=300$). External judges form a coherent block (0.63–0.75); the in-family judge barely correlates with any of them (0.27–0.31)—it is measuring something related but different.

Frontier models on TRIZ-Bench

We ran GPT-5.4, Claude Sonnet 4.6, Claude Opus 4.8, and Gemini 3.5 Flash as *contestants* under the identical protocol (5,983 valid scores; seven item-level scores lost to API safety refusals and treated as missing, never zero). Table 4 condenses the full per-judge matrix.

TRIZ-Bench contestant table. kw: keyword hit rate. Judge cells: overall mean (paired diff vs. base [95% CI]); “—” marks family recusal.

Contestant	kw	Claude	Gemini	GPT	Moonshot
base	0.638	2.555	2.871	2.301	3.030
Meerkat-TRIZ v1	0.638	2.649 (+0.09*)	2.823 (−0.05)	2.405 (+0.10*)	3.423 (+0.39*)
GPT-5.4	0.628	2.692 (+0.14*)	3.094 (+0.23*)	—	3.213 (+0.18*)
Sonnet 4.6	0.576	—	2.846 (−0.04)	2.299 (−0.02)	3.080 (+0.05)
Opus 4.8	0.576	—	2.776 (−0.10)	2.204 (−0.10)	2.572 (−0.46*)
Gemini 3.5F	0.599	2.863 (+0.31*)	—	2.587 (+0.29*)	3.268 (+0.23*)

Three findings stand out. First, on the *objective* track the 35B base and its fine-tune beat every frontier model we tested—domain keyword coverage is where specialization shows up with certainty. Second, on the judge track the strongest frontier models (Gemini, GPT-5.4) still lead the fine-tune; their answers are longer and more elaborately structured, and judges—even with an anti-verbosity rubric—reward some of that. We read the residual gap as partly real (elaboration quality) and partly the known length confound (Section 7); v5b’s targeted injection attacked the real part and was nonetheless rejected by the gates (Section 5.4). Third, Claude Opus 4.8 scores *below* the 35B base on every judge that is allowed to score it, and significantly so on the in-family judge (−0.458 [−0.575, −0.344])—a reminder that frontier general models do not automatically dominate narrow methodological domains, and a result we could not have credibly published without the recusal-and-disclosure protocol.

General ability is not sacrificed

On the 120-item general probe: base 84.03%, predecessor (v2) 86.81%, Meerkat-TRIZ v1 85.56%. The candidate-vs-predecessor difference of −1.25pp is far above the G6 floor (−5pp), and the candidate sits *above* base. Domain fine-tuning at 0.24% of parameters did not buy domain skill with general regression.

The v5b case study: a correctly rejected repair

Section 6 attributes the previous generation’s principle-recommendation weakness to three data defects. v5b tested the repair with a single-variable change: 408 injected commit-style items (length band 800–1600 characters, all 40 inventive principles covered, gold-decontaminated) plus removal of the 83 mislabeled items, for 11,421 training pairs; every other subset was byte-identical to v5a and every hyperparameter was frozen.

The repair worked *behaviorally* and failed *measurably*. Qualitatively, v5b no longer hedges: sampled principle-recommendation answers now commit to numbered principles with application logic (“Principle one: segmentation...”), exactly the deficit the attribution identified. Quantitatively, the gates rejected the generation on two independent significant track-reversals (Table 5). Against base, the in-family judge-track gain grew to +0.430 while the objective track went significantly *negative* (−0.028); against v2 on the targeted subset the same flip appeared in extreme form (judge +0.800 vs. keyword −0.135). G7 fired. The external panel made the rejection unambiguous: all three external judges scored v5b statistically indistinguishable from base (+0.067/−0.003/+0.015, all n.s.), so unlike v5a—whose small gain two of three external judges confirmed—v5b’s larger in-family headline has *zero* external confirmation.

v5b dual-track outcome (paired diffs, 95% CI). The two significant reversals that fired G7 are marked †.

Comparison	judge track	keyword track
v5b — base, overall†	+0.430 [+0.323, +0.537]	−0.028 [−0.048, −0.009]
v5b — base, principle rec.	−0.033 [−0.217, +0.167]	−0.104 [−0.159, −0.049]
v5b — v2, principle rec.†	+0.800 [+0.617, +1.000]	−0.135 [−0.193, −0.082]
External judges (v5b — base)	claude +0.067 n.s.; gpt −0.003 n.s.; gemini +0.015 n.s.	

Attribution of the keyword regression found it to be primarily a measurement artifact rather than degraded principle selection. Of 59 lost keyword instances on the 60-item subset, only 22% are principle-name mismatches (v5b recommending, e.g., segmentation where the reference chose prior action); 78% are generic framing tokens (*TRIZ*, *inventive principle*, *technical contradiction*) that base’s longer answers sprinkle and the terse commit-style format omits. v5b in fact hits expected keywords at *higher* density than base (1.88 vs. 1.03 per 1,000 characters); its absolute hit count falls because its answers are 0.39× as long (median 1,055 vs. 2,792 characters), and a counterfactual holding base’s density fixed attributes the entire net drop to this length surface. The objective track thus measures a length-sensitive vocabulary contract rather than what the injection changed; meanwhile all judges, external included, scored the two arms equal, so no judged-quality gain offsets the contract loss. The decision gates’ verdict—G7 fail, G3 freeze pending formal attribution—is that this trade is not worth making: v5b is rejected, and Meerkat-TRIZ v1 (v5a) remains the shipped model. We report v5b in detail precisely because a single-metric

pipeline would have shipped it: its in-family headline (+0.430) is the largest in the lineage. Section 5.5 later shows that both of v5b’s significant tracks against base decompose into length-surface effects.

The equal-length control

Section 5.1 leaves one confound standing: base answers are much longer than v5a’s (median 3,242 vs. 1,344 characters), and the in-family rubric explicitly penalizes verbosity. If the rubric works, part of v5a’s +0.393 could be a length artifact rather than a quality gain. An earlier attempt to isolate the effect by score-bucket pairing was underpowered (17/300 same-bucket pairs), so we ran a direct control: base regenerated all 300 items with a per-item length target equal to v5a’s answer length on that item (“keep the answer to about N characters $\pm 10\%$ ”), every other protocol element byte-identical; judging and inference are unchanged.

Length compliance was partial. The instruction asked for a median of 1,344 characters; base produced a median of 1,750 ($1.29\times$ overshoot; 20/300 items inside the $\pm 10\%$ band), compressing the base-to-v5a length ratio from $2.4\times$ to $1.3\times$. We report the control as-is rather than iterating toward perfect compliance.

Equal-length control, in-family judge (0–4), paired diffs with 95% CIs (10,000 paired bootstraps, seed 42). Row 1 recomputes Table 2 on the control’s paired sample and reproduces it exactly; rows 4–5 apply the same control to the rejected v5b generation at zero additional generation cost.

Comparison	diff [95% CI]
v5a — base (unconstrained)	+0.393 [+0.297, +0.490]
v5a — base (length-matched)	−0.070 [−0.140, +0.000]
base (matched) — base (unconstrained)	+0.463 [+0.367, +0.557]
v5b — base (unconstrained)	+0.430 [+0.323, +0.537]
v5b — base (length-matched)	−0.033 [−0.110, +0.043]

Table 6 separates artifact from gain for the in-family judge, and two conclusions stand. First, **v5a’s in-family advantage does not survive length control**: with base constrained toward v5a’s length the paired difference is −0.070 and statistically null —the +0.393 headline is predominantly a length artifact. Mean scores make the mechanism visible: base 3.030 unconstrained, 3.493 length-matched, v5a 3.423. Second, **the anti-verbosity rubric is real**: base’s only change was a length instruction, and its judged score rose by +0.463—the judge penalizes verbosity itself, not capability.

The equal-length control under *external* judges (v5a — base, paired diffs [95% CI], $n=299$). Matched-arm CIs are additionally widened by judge rerun-noise synthesis (Section 7, rule D6): Claude [+0.105, +0.270], GPT [+0.161, +0.380]—both remain significant.

Judge	unconstrained	length-matched
Claude Sonnet 4.6	+0.094 [+0.020, +0.167]	+0.187 [+0.110, +0.264]

Judge	unconstrained	length-matched
GPT-5.4	+0.104 [+0.020, +0.184]	+0.271 [+0.177, +0.368]
Gemini 3.5 Flash	−0.048 [−0.144, +0.045]	+0.025 [−0.074, +0.120]

Third, **the external judges reverse the picture—upward** (Table 7). Scoring the same matched-length arm, two of three external judges rate v5a significantly *above* base. The mechanism mirrors the in-family one with the sign flipped: external judges *lower* base’s score when it is forced to write short (GPT 2.301 → 2.134; Claude 2.555 → 2.462)—they reward elaboration, so base’s unconstrained sprawl was itself buying judged quality. At a level length budget the fine-tune’s content is genuinely better; base closes most of the gap only by spending $2.4\times$ the tokens, and even then not all of it (the unconstrained residuals +0.09/+0.10 of Section 5.1). The four readings now compose a single consistent picture: in-family +0.39 (anti-verbosity over-credits) → in-family matched −0.07 (rubric real but conservative) → external unconstrained +0.09/+0.10 (length and content offset) → external matched +0.19/+0.27 (content quality, isolated).

The keyword track sharpens rather than contradicts this: unconstrained, v5a and base are tied (−0.0001 n.s.), but length-matched base *loses* keyword coverage relative to its unconstrained self (−0.023 [−0.040, −0.005]) while v5a does not—so at matched length v5a significantly *out-covers* base on keywords (+0.023 [+0.005, +0.041]). Brevity costs the base keyword surface that the fine-tune retains at $0.4\times$ the length. The same control applied to the rejected v5b generation (Table 6, rows 4–5) dissolves *both* of its significant tracks: the judge gain (+0.430 → −0.033 n.s.) and the keyword loss (−0.028 → −0.006 n.s.) are alike length-surface effects—v5b’s behavioral change is real, but every measured effect against base was a length phenomenon (its regression against v2 on the targeted subset is not length-controlled here and stands).

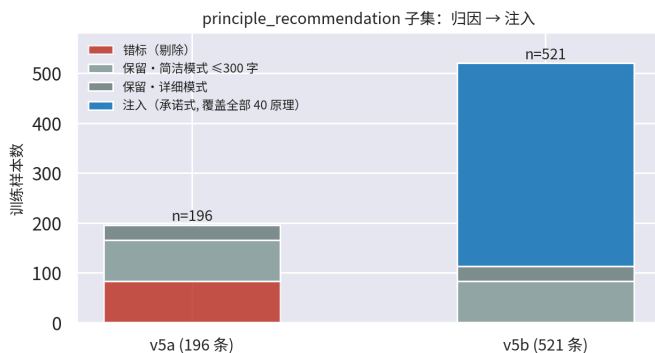
The claim this paper makes is therefore stronger than parity: at matched length the fine-tune outperforms its base under two of three external judges, and at natural lengths it delivers that content at $0.4\times$ the answer length with keyword coverage the base cannot retain when compressed. We now require an equal-length control as a mandatory companion to any judge-track significance claim (harness rule D5), external panel included.

Analysis

What was actually wrong with principle recommendation

The previous generation’s weakest subset illustrates why measurement must precede data work. Four judges agreed the fine-tune did not beat base on principle recommendation (+0.05, +0.05, −0.08 n.s.; −0.31 significant negative). Attribution found three independent defects: (i) *quantity*—196 training items, the smallest domain subset; (ii) *style*—72% of them concise-mode items capped at 300 characters, teaching terseness on a task whose gold answers argue; (iii) *labels*—42%

mislabeled, including ARIZ resource questions and standard-solution questions. Qualitatively, the model had learned to answer “recommend two principles” prompts with a *methodology lecture* (“step one, restate the contradiction; step two, consult the matrix...”), hedging instead of committing, while the base model committed to numbered principles with application logic and was rewarded for it. The keyword track being significantly *positive* (+0.049) on the same subset proves the model knew the vocabulary; what it lacked was the behavior, because the behavior existed in only 54 training examples (Figure 4).



Principle-recommendation subset before and after attribution-driven injection. Left: the v5a subset (196 items) decomposed by the audit—83 mislabeled items removed, 83 concise-mode and 30 detailed legacy items kept. Right: the v5b subset (521 items) adds 408 injected commit-style items covering all 40 inventive principles. All other subsets are byte-identical between v5a and v5b.

Lessons the gates enforced

Three rejected generations anchor the methodology’s value. v3 pushed keyword-rich data: the objective track improved and the judge track significantly regressed—the G7 circuit breaker fired, and attribution showed keyword stuffing had degraded answer quality. v4 rebalanced data and quietly dropped eight expected keywords in concept explanation (true regression, not artifact): G3 fired. v5b repaired a documented behavioral deficit and produced the lineage’s largest in-family headline (+0.430)—and was rejected when the objective track flipped sign on two comparisons and no external judge confirmed the gain (Section 5.4). All three rejections were of generations that single-metric evaluation would have shipped. The counter-discipline also matters: frozen hyperparameters made every comparison single-variable, and the decision to never “tune the learning rate” when a data fix was available is why the lineage reads as a sequence of interpretable experiments rather than a soup.

Judge bias, mapped

Our measurements localize four judge biases in this domain: *family inflation* ($\sim 4\times$ effect-size deflation when moving from in-family to external judges); *leniency offset* (Gemini scores all arms $\sim 0.3\text{--}0.6$ higher than GPT while agreeing on rankings, $\rho = 0.69$); *elaboration preference* (frontier models’ judge-track advantage co-occurring with $2\text{--}3\times$ answer length despite an anti-verbosity rubric); and, newly evidenced by

v5b, *data-generator–judge family alignment*: v5b’s injected training data was generated by a model of the same family as the in-family judge, and its in-family headline *grew* (+0.430 vs. v5a’s +0.393) while external confirmation collapsed to zero (+0.067/−0.003/+0.015 n.s.). Family bias thus propagates through the data chain, not only the generation chain: when an intervention’s data comes from family *X*, a family-*X* judge is doubly compromised, and only the external panel is insulated. The first two biases are handled by recusal plus disclosure; the third is now resolved by the equal-length control (Section 5.5), which shows the in-family rubric’s verbosity penalty is genuine, that the fine-tune’s in-family advantage is predominantly a length artifact—and, symmetrically, that external judges *lower* the length-matched base (GPT 2.301 → 2.134), a direct measurement of elaboration preference that isolates the fine-tune’s content-quality gain (+0.19/+0.27); the fourth is handled by treating any same-family intervention as automatically suspect under G8.

Limitations

(i) The judge-track effect is judge-dependent; the length-matched external spread (+0.03 to +0.27) is our best estimate of the true content-quality effect, but it inherits each judge’s rubric idiosyncrasies. Relatedly, the external judges—served through an API gateway—are *not* deterministic at temperature 0 (per-item flip rates 0.18–0.80 over three repeated runs of 50 items; the in-family judge, called directly, had flip rate 0.000). Published external differences were computed without judge rerun variance; re-widening the CIs by rerun-noise synthesis leaves the significant results significant (+0.104 → [+0.006, +0.202]; +0.094 → [+0.015, +0.173]; matched-arm +0.187/+0.271 likewise survive), but single-run external numbers should be read with this noise budget in mind (harness rule D6 now gates judge determinism). (ii) The length confound on the judge track is bounded but not perfectly removed: the equal-length control (Section 5.5) overshot its per-item targets by 29% (median 1,750 vs. 1,344 characters), and a stricter-compliance replication is future work. (iii) The benchmark is Chinese-only and TRIZ-specific; nothing here tests cross-lingual or cross-domain transfer. (iv) Public benchmark questions derive from copyrighted source texts (model-generated derivatives); we publish questions and keywords but not reference answers, and a clean-room v2 benchmark plus an English subset is planned. (v) Human-expert validation of judge reliability is scheduled (30-item blind paired study) but not yet complete at submission time; judge–human agreement is therefore asserted by proxy (inter-judge agreement), not measured. (vi) A single base model and a single adapter configuration are tested; routed-expert coverage and rank ablations are open engineering questions. (vii) v5b, the most recent generation and the only attempted data-level repair, was rejected by the gates (Section 5.4): how to fix principle recommendation in a way that satisfies both tracks remains an open problem. Preference optimization on judge-verified pairs (e.g., DPO) is a natural next step but is deliberately deferred until judge reliability is human-validated, to avoid distilling a judge’s idiosyncrasies into weights.

Conclusion

We set out to answer whether a TRIZ-domain fine-tune is real, and built the measurement apparatus required to make the answer checkable. The answer is: real on the objective track, where the fine-tune matches its base and both beat every frontier model we tested; real as judged quality *efficiency*—at natural lengths it delivers its content at $0.4\times$ the base’s answer length—and, once length is controlled, real as a judged-quality *gain*: two of three external judges score the fine-tune significantly above the length-matched base ($+0.19/+0.27$), even though the in-family headline itself proved to be a length artifact (Section 5.5); achieved without general-capability sacrifice; and improving through a gated, failure-driven data lineage rather than through training heroics. We release model, harness, benchmark, and caches. If the field adopts one habit from this paper, we hope it is this: never let the family that generated the data be the only judge of the model—and never print a judge-track gain without a length control.

Generation and judging configurations

All contestant generations use the system message: “You are a TRIZ innovation-methodology expert assistant; answer professionally in Chinese on TRIZ theory, inventive principles, contradiction analysis and ARIZ” (translated), max 2048 tokens. Judges use the Arm-A rubric with five 0–4 dimensions, batch size 5, temperature 0, full untruncated input; batch parse failures degrade to single-item retries; API safety refusals are logged as missing. All bootstrap procedures use `random.Random(42)` with 10,000 resamples. Judge caches for every $\text{arm}\times\text{judge}$ cell are published alongside the harness.

Gate definitions (machine-checkable)

G0: refusal-template hit rate $\leq 2\%$. G1: vs-base Arm-A CI lower bound > -0.15 (Arm B must not contradict). G2: vs-predecessor CI lower bound > -0.05 and (judge or keyword significantly positive). G3: any subset keyword CI upper bound < 0 requires attribution; true-missing $\Rightarrow$ FAIL, artifact $\Rightarrow$ judge-track arbitration. G4: $n < 30$ subsets descriptive only. G5: ARIZ track divergence resolved for the rubric track. G6: probe diff $> -5\text{pp}$. G7: any dimension with both tracks significant in opposite directions $\Rightarrow$ FAIL. G8: missing external review $\Rightarrow$ SKIP; in-family significantly positive with all-external significantly negative $\Rightarrow$ FAIL; majority-external significantly negative $\Rightarrow$ FAIL.

99

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